

21.1: Using Differentiation

$\vec{v} = \frac{\Delta s}{\Delta t}$ $\quad \vec{a} = \frac{\Delta \vec{v}}{\Delta t}$

$\Rightarrow v = \frac{ds}{dt}$ $\quad \Rightarrow \vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2 s}{dt^2}$

These functions find gradients

Using integration:

For general functions of velocity & position (displacement):

$v(t) = \int a(t)\,dt$, $s(t) = \int v(t)\,dt$. These may have constants added, be vigilant.

To find a quantity, use definite integration

$\Delta v = \int_{T_2}^{T_1} a(t)\,dt$, $\Delta x = \int_{T_2}^{T_1} v(t)\,dt$

These Techniques can be used to derive SUVAT.

$\int a\,dt = at + C = v = u + at$.

$\int (u + at)\,dt = C + ut + \frac{1}{2} at^2 = s = s_0 + ut + \frac{1}{2} at^2$

$\Rightarrow s = \sum_{n=0}^{\infty} \frac{s^{(n)}(0)}{n!} t^n$ (an approximation at $t = 0$)